MOCHAMMAD IRHAM MAULANA

Informatics Engineering Student | Front-End & Software Engineer

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PROFESSIONAL SUMMARY

Adaptable and results-driven Informatics Engineering student at Institut Teknologi Sepuluh Nopember (ITS) with practical experience in Front-End Web Development, Secure Software Development Life Cycle (SSDLC) framework formulation, and Software/Game Engineering. Proven track record in developing high-performance web systems, establishing enterprise-grade application security standards, and engineering interactive software architectures. Strong communicator and collaborative team leader with demonstrated commitment to clean code, usability, and robust system integrity.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) 2022 – 2026
Bachelor of Engineering in Informatics (Teknik Informatika) Surabaya, Indonesia

- **Relevant Coursework:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Web Development, Database Management Systems, Software Engineering, Network & Computer Security.

PROFESSIONAL EXPERIENCE

PT PLN Indonesia Power September 2025 – November 2025
Software Security & SSDLC Intern Surabaya, Indonesia

- Formulated and developed comprehensive Secure Software Development Life Cycle (SSDLC) guidelines and documentation to establish robust security baselines for enterprise application development.
- Analyzed end-to-end software development workflows, identified security risks, and integrated security checkpoints—including threat modeling, secure coding standards, and vulnerability assessment protocols—across each lifecycle phase.
- Designed practical implementation guides and compliance checklists enabling engineering teams to build secure, reliable, and audit-ready digital solutions.

PT Trans Pacific Petrochemical Indotama (TPPI) July 2025 – August 2025
Front-End Developer Intern Tuban, Indonesia

- Engineered, styled, and maintained responsive web applications and internal operational systems supporting day-to-day enterprise and industrial workflows.
- Translated complex operational requirements into intuitive, accessible, and high-performance user interfaces with clean cross-browser compatibility.
- Built modular and reusable UI components, optimized client-side asset loading, and collaborated with cross-functional teams to integrate APIs and maintain consistent design standards.

PROJECTS

Petani Sawit – Agricultural Simulation Game Unity, C#, URP

- Engineered a time-based agricultural resource management game modeling the 25-year lifecycle of oil palm plantations across 3 biological growth phases.
- Implemented dynamic monthly commodity market pricing (±20% fluctuation), a bank loan system with interest modeling, and event-driven risk-management mechanics.
- Architected pure C# simulation logic (`GameStateLogic`) following Separation of Concerns (SoC) principles decoupled from Unity engine components, integrated with JSON data persistence.

TPPI Web Application – Operational System HTML5, CSS3, JavaScript, PHP, SQL

- Developed interactive internal operational dashboards facilitating data presentation and task monitoring for corporate personnel.
- Designed modern, user-friendly UI layouts optimized for responsiveness across mobile, tablet, and desktop environments.

Math Blaster – Custom Raycaster Graphics Engine JavaScript, HTML5 Canvas

- Engineered a retro 3D pseudo-FPS raycaster rendering engine from the ground up using pure Vanilla JavaScript and HTML5 Canvas without external rendering frameworks.
- Implemented real-time linear algebra transformations, Digital Differential Analysis (DDA) raycasting algorithms, texture mapping, and collision detection.

Desktop Performance & System Utilities (Game Optimizer, Center Window, SD App) C#, .NET, Python, Win32 API

- Created Windows desktop utilities using C# WinForms/WPF and Python for process optimization, window centering via Win32 API, and precision shutdown automation.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

HMTC ITS (Informatics Student Association) 2024
Internal Affairs Staff Surabaya, Indonesia

- Coordinated internal student engagement and relationship management across academic cohorts, overseeing department-wide events and community initiatives.

Schematics ITS (National Technology Event) 2022 – 2024
Public Relations Expert Staff (2024) | PR Staff (2023) | Volunteer Staff (2022) Surabaya, Indonesia

- Led external communication and corporate partnership acquisitions, pitching event value propositions and managing high-value relationships with external corporate sponsors.

TECHNICAL SKILLS & COMPETENCIES

Programming Languages: C, C++, C#, Python, JavaScript, PHP, SQL, HTML5, CSS3

Frameworks & Libraries: Unity (URP), .NET (WPF, WinForms), Ren'Py, Vanilla JS / Canvas API

Developer Tools & Platforms: Git, GitHub, PostgreSQL, Android Studio, Blender, Linux, Windows, Microsoft Office

Areas of Expertise: Secure Software Development Life Cycle (SSDLC), Front-End Engineering, UI/UX Principles, Object-Oriented Programming (OOP), Database Systems, Game Architecture

Languages: Indonesian (Native), English (Professional Working Proficiency)